Name: ______________________________ **Date:** _________________ **Class:** _______________

MyDesign Portfolio

Element L Initial Template

Element L: Presentation of Designer's Recommendations

Content of Recommendations *(L1, L2)*

Earlier on (in K3, if you completed it) you provided general lessons learned related to engineering design, and now (in L2) you are asked to provide recommendations for continuing work on this project in particular.

Answer as many of these questions as you can with specific recommendations based on your testing data, expert/stakeholder feedback, and any new research you've gathered. These recommendations may include process improvements (how you worked) and/or project improvements (actual design specific decisions you made). Provide recommendations to someone doing a similar project that might build or continue upon your work. For example: "Taking into account <something we did>, we believe doing _________ would improve _________ because ________."

- What could be done next to go further?
- What would you keep if you did the project again?
- What would you change?
- What would have made this project easier for you to complete?

In order to earn 5's on Element L, you will need specific and detailed ways the project can be improved and a rationale for why they are needed.

- What could be done next to go further?

The main problem that I felt we had with the ROV was in the bay the movement of the water was strong enough that it was a little difficult to move such a light ROV around. One simple change would be a more massive ROV with more powerful motors. Also we found that samples could get washed out if we turned or went backwards. This could be solved by some system that doesn't allow water to flow back through the 5mm filter. Additionally, stability in the water, which we solved by adjusting the buoyancy, could be solved with more degrees of freedom and a gyroscope, although that's a significant upgrade.

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- What would you keep if you did the project again?

Given similar materials we'd want to keep the motor layout and frame. Despite some troubles, we found our robot to maneuver very well, and although it could be faster, we were generally able to get the robot to any position and orientation that we wanted. Also building in a way to access the collected samples worked very well. We'd want to keep a similar system.

- What would you change?

We'd want to rethink the sample collection system. I don't have the full context for how well it worked, but I know that it would work better if we eliminated the problem with how samples could flush out of the collector. That does likely mean the design will be a lot more complicated.

- What would have made this project easier for you to complete?

Being more accurate with frame measurements. PVC pipe cutters aren't the best and result in really sloped cuts. Using a sort of saw, whether that be a power tool or not would produce a much more symmetric frame.

Detailed Plans for Implementation of Improvements *(L2)*

Now that you have generated a list of specific recommendations with their rationale, go back under each and generate a consistently detailed list about HOW these details will be implemented. Refer back to Element I where you concluded with some of these ideas.

- More massive and powerful

I don't think changing the mass of the ROV will produce as meaningful of a result as changing the motors. The first step would be to look at similar models for underwater motors, to see if there's an upgrade that's needed. More powerful motors will definitely need 12v, and likely also require us to step up the breaker to something as high as (maybe?) 30 amps.

- More symmetric frame

Creating a more thoughtful cut list, meaning listing out all the lengths of PVC pipes needed. Then making sure the same lengths are cut together, marking them, and then in the best case we'd have some sort of chop saw like the one in the garage which cuts straight up and

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down, but it's also reasonable to be more mindful with the PVC cutters, and make sure that you stay on the marked line and rotate while cutting.

- Rethought sample collection

I think that this would be significantly more complex if we wanted to filter out the <1mm microplastics. One question to ask to Mary is if filtering out junk <1mm is more important than retaining all samples that flow through our ROV.

This is a very broad idea for a design. With the left being the back and the right being the front (up being the top, down being the bottom). This is a flat 2d view of a slice of the design.

Generally water flows from right to left, samples pass through the 5mm, the water pushes the latch open, and then samples get stuck at the back 1 mm filter. The front is also sloped to encourage samples stuck at the 1mm filter to flow into the 5mm filter.

The latch is also on an elastic band, so that when we slow down, it shuts, and when we go backwards, water keeps it pushed down, and all our samples slosh back to the 1mm end, where they are secure.